

Lexicographic EU

Model — human; produced by literature.

0.1 Package

- **Completeness.** For all $p, q \in \Delta(X)$: $p \succcurlyeq q$ or $q \succcurlyeq p$.
- **Transitivity.** For all $p, q, r \in \Delta(X)$: if $p \succcurlyeq q$ and $q \succcurlyeq r$ then $p \succcurlyeq r$.
- **Independence.** For all $p, q, r \in \Delta(X)$ and $\lambda \in (0,1]$: $p \succcurlyeq q$ if and only if $\lambda p + (1-\lambda)r \succcurlyeq \lambda q + (1-\lambda)r$.
- **$\neg$ Mixture continuity.** For all $p, q, r \in \Delta(X)$ the sets $\{\lambda \in [0,1] : \lambda p + (1-\lambda)q \succcurlyeq r\}$ and $\{\lambda \in [0,1] : r \succcurlyeq \lambda p + (1-\lambda)q\}$ are closed.
- **$\neg$ Archimedean.** For all $p, q, r \in \Delta(X)$: if $p \succ q \succ r$ then there are $\alpha, \beta \in (0,1)$ with $\alpha p + (1-\alpha)r \succ q$ and $q \succ \beta p + (1-\beta)r$.

0.2 Construction

Lexicographic expected utility. With $X = \{a, b, c\}$ let $p \succcurlyeq q$ iff $p(a) > q(a)$, or $p(a) = q(a)$ and $p(b) \geq q(b)$. This is complete and transitive (a lexicographic order on $\mathbb{R}^2$), and satisfies independence because mixing with r multiplies the vector of differences by λ , which preserves lexicographic sign. But $\{\lambda : \lambda \delta_a + (1-\lambda)\delta_c \succcurlyeq \delta_b\} = (0, 1]$, which is not closed.

The same lexicographic order ($p \succcurlyeq q$ iff $p(a) > q(a)$, or $p(a) = q(a)$ and $p(b) \geq q(b)$). Here $\delta_a \succ \delta_b \succ \delta_c$, but every mixture $\beta \delta_a + (1-\beta)\delta_c$ with $\beta > 0$ puts positive weight on a and so is strictly preferred to δ_b ; no β with $\delta_b \succ \beta \delta_a + (1-\beta)\delta_c$ exists.

0.3 Notes

Derivable from lexicographic-not-archimedean together with continuity-completeness-imply-archimedean (the validator reports it as redundant); kept because it is the form the literature states.

0.4 Sources

- Hausner (1954), Multidimensional utilities, in Decision Processes (Thrall, Coombs & Davis eds.)
- Fishburn (1971), A study of lexicographic expected utility, Management Science 17
- Hausner (1954), Multidimensional utilities
- Fishburn (1971), A study of lexicographic expected utility, Management Science 17

Record: topics/decision-theory/models/lexicographic-eu.yaml